

CS 310: Assignment 1

Instructions

- Answer all questions clearly and concisely.
 - Submit a single handwritten PDF via Moodle before the deadline. Eg. 23B0000.pdf
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Question 1.

Construct a deterministic finite automaton (DFA) for the language

$$L = \{w \in \{0, 1\}^* \mid n_0(w)^3 + n_1(w) \equiv 0 \pmod{5}\}$$

where $n_0(w)$ denotes the number of 0s in the string w , and $n_1(w)$ denotes the number of 1s. Clearly describe the states, transitions, and acceptance conditions of your DFA.

Question 2.

Design a deterministic finite automaton (DFA) for the language

$$L = \{w \in \{a, b\}^* \mid \# \text{occurrences of } \mathbf{ab} \text{ in } w = \# \text{occurrences of } \mathbf{ba} \text{ in } w\}.$$

That is, the string should contain an equal number of substrings **ab** and **ba**.

Question 3.

Consider the following language:

$$L = \{w \in \Sigma^* \mid w = uv, \text{ where } u, v \in \Sigma^*, n_0(u) \equiv 0 \pmod{2}, n_1(v) \equiv 0 \pmod{2}\}.$$

This language consists of all strings that can be split into two parts u and v , such that the number of 0s in u is even, and the number of 1s in v is even. You may assume $\Sigma = \{0, 1\}$. Design a DFA that recognizes this language, clearly explaining its states and transitions.

Question 4.

Write regular expressions for the following languages:

- (a) The set of all strings over $\{0, 1\}$ such that every pair of adjacent 0s appears before any pair of adjacent 1s. That is, no substring **11** occurs before a substring **00** in the string.
- (b) The set of all strings over $\{a, b\}$ in which no two *b*s occur adjacent to each other. That is, the substring **bb** is not allowed anywhere in the string.
- (c) For the complement of the language described in question 3.

Question 5.

Describe the languages represented by the following regular expressions. Then, determine whether the two regular expressions define the same language. Justify your answer.

(1) $r_1 = (01)^* + (10)^* + 0(10)^* + 1(01)^*$

(2) $r_2 = (\varepsilon + 1)(01)^*(\varepsilon + 0)$

Question 6.

Write context-free grammars (CFGs) for the following languages:

- (1) $L = \{a^i b^j c^k \mid i \neq j \text{ or } j = k\}$
- (2) The language of **unbalanced parentheses**, where the symbol *a* represents an opening parenthesis (, and *b* represents a closing parenthesis).

A **balanced** parenthesis string (in terms of *a* and *b*) is defined as any string where:

- Every opening parenthesis *a* is matched by a corresponding closing parenthesis *b*, and
- Parentheses are properly nested.

Your task is to write a CFG for the language of strings over $\{a, b\}$ that are **not** balanced. That is, the complement of the balanced parentheses language.

Question 7.

Construct a Type 0 grammar to generate a^n where *n* is a perfect square.